

Oncogenetics

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Objectives

1-Differentiation between genetic and environmental factors in cancer

2-Inherited cancer syndromes:

- ❑ HNPCC (Lynch syndrome)

- ❑ Familial adenomatous polyposis (FAP)

- ❑ Breast cancer

3-Genetic counseling in familial cancer

4-Screening of familial cancer

5-Explain methods of prevention or early detection of familial cancer

Differentiation Between Genetic and Environmental Factors in Cancer

Is Cancer a Genetic Disease?

- It is important to understand that all cancer is a genetic disease due to:
 - Aberrant in cell division
 - Loss of normal apoptosis

BUT not all cancers are inherited.....

- For most types of cancer, the mutations that occur in the genes of the cancer cell are caused by exposures to environmental factors that are controllable.

- **Inherited cancers:**

Some people is strongly predisposed by inherited germline pathogenic variants.

- **Example:** breast, ovarian, colon.....
- Genetic testing is now available for some hereditary cancers.

Does the Cancer Gene Skip Generations?

Yes.....

Why?

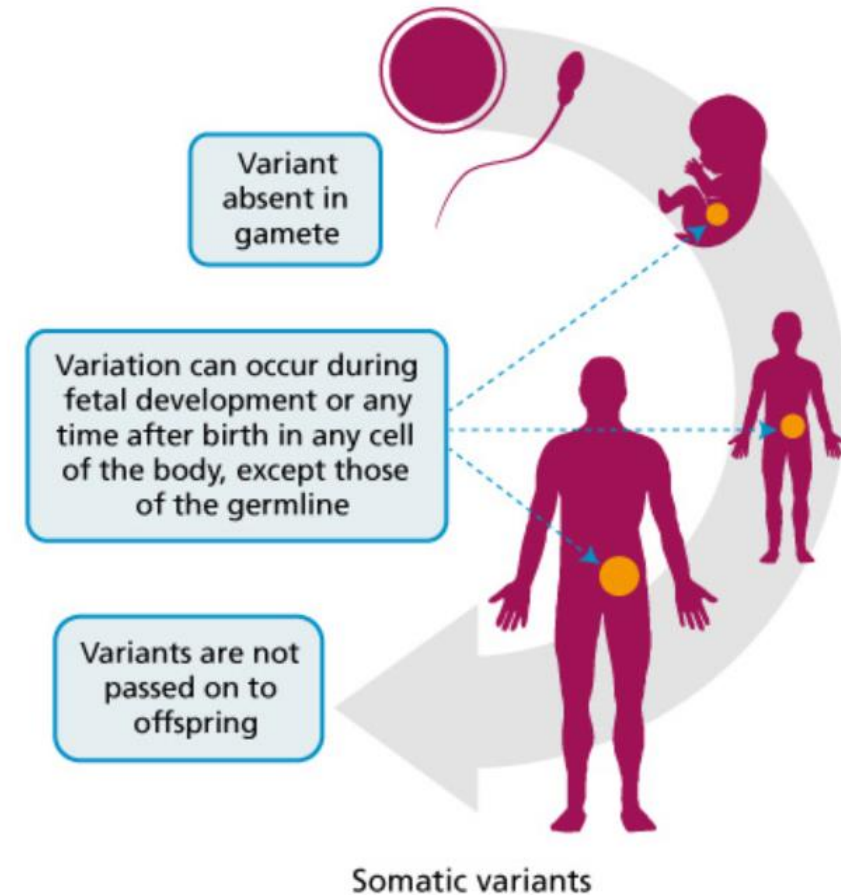
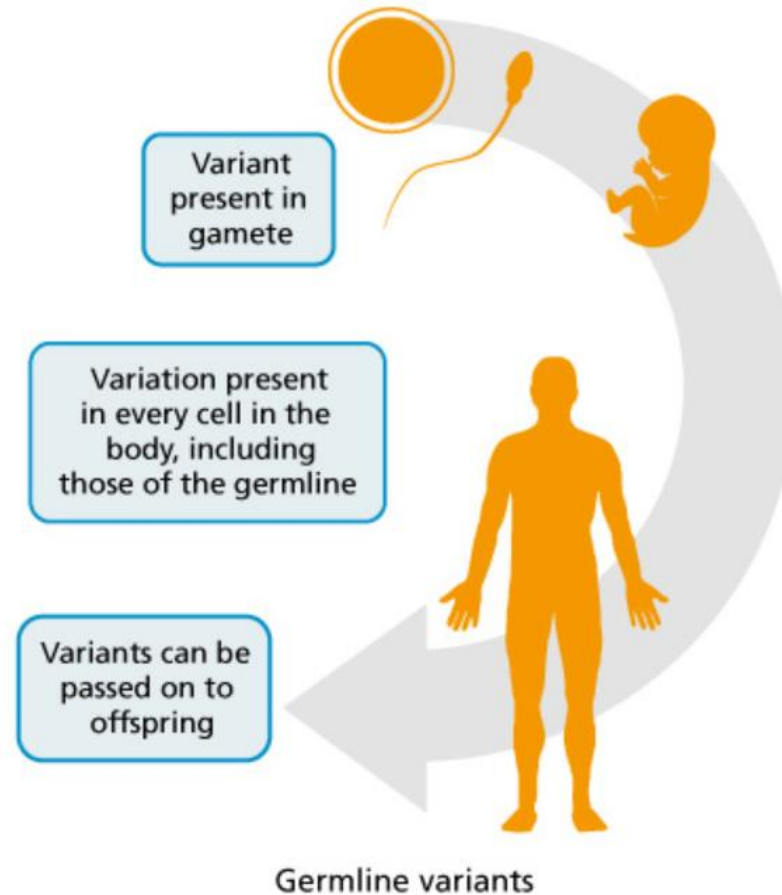
A parent may have the gene and not develop cancer **But** his child who inherits the same gene does develop cancer (Why? → Penetrance,,,,, two-hit hypothesis).

Can Cancer Run in Families?

Genetics

Germline vs Somatic Mutations

Germline mutation can be inherited but somatic one cannot be inherited.



Cancer Predisposition Genes

- **Oncogenes**

Oncogenes are the altered forms of normal genes (protooncogenes) that have key roles in cell growth and differentiation pathways.

- **Tumor Suppressor Genes**

The study of human hereditary cancer has identified the existence of tumor suppressor genes, which control the cell proliferation (suppress inappropriate cell proliferation) and constitute the largest group of hereditary cancer genes.

- **DNA mismatch repair genes**

which are responsible for the detection and repair of genetic damage within the cell.

Family Studies

- The clustering of cancer cases within families can provide significant evidence supporting a genetic contribution.
- For example: a woman who has a first-degree relative with breast cancer the risk that she will also develop breast cancer is between 1.5 and 3 times the risk for the general population.
- The earlier the age at diagnosis, the greater the risk to close relatives (table 14.7).

Genetic Architecture of Cancer Risk

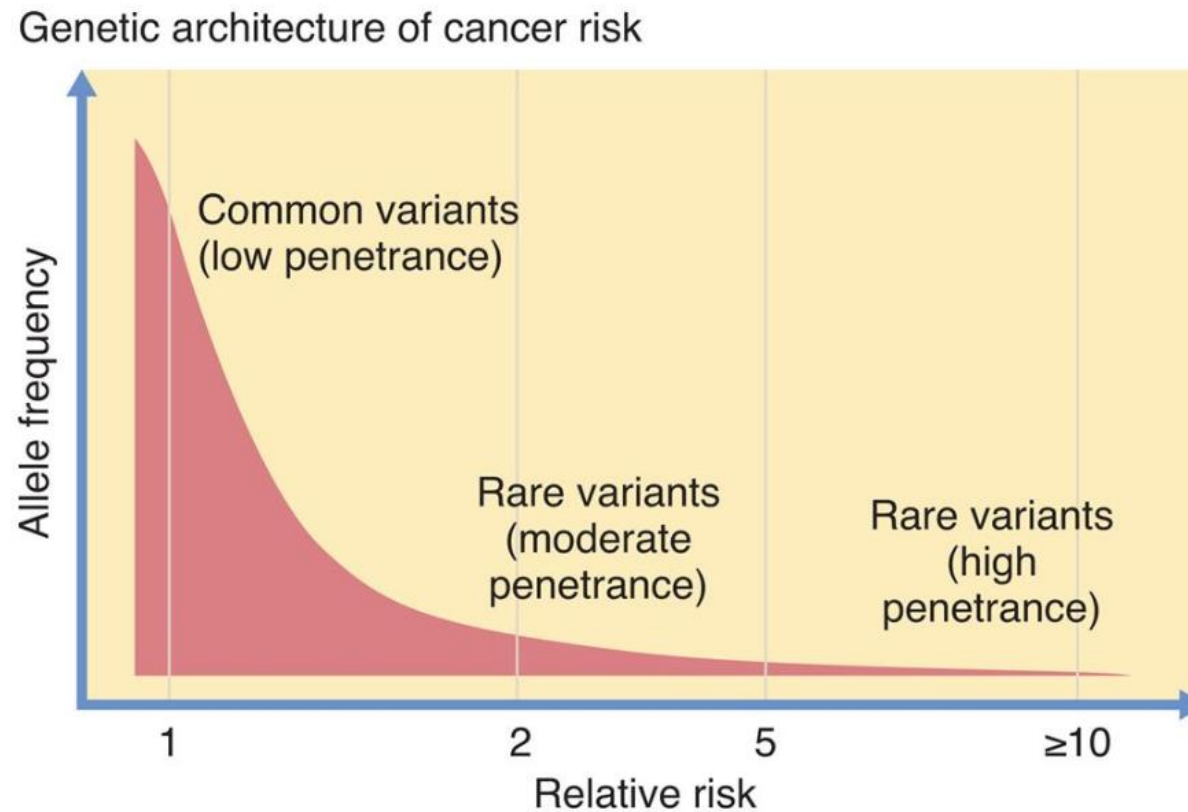


FIG. 14.2 Genetic architecture of cancer risk. In the majority of cancers there is a low risk to relatives because genetic associations tend to be with common, low-penetrance alleles identified in genome-wide association studies. The risk to relatives increases markedly when the cancer is associated with rare, high-penetrant genetic variants, such as variants in the *BRCA1/BRCA2* genes, e.g., hereditary breast and ovarian cancer, and in the mismatch repair genes associated with Lynch syndrome. From National Cancer Institute: PDQ Cancer Genetics Overview. Bethesda, MD: National Cancer Institute. Date last modified January 15, 2016. Available at: <http://www.cancer.gov/about-cancer/causes-prevention/genetics/overview-pdq>. Accessed August 2020. With permission.

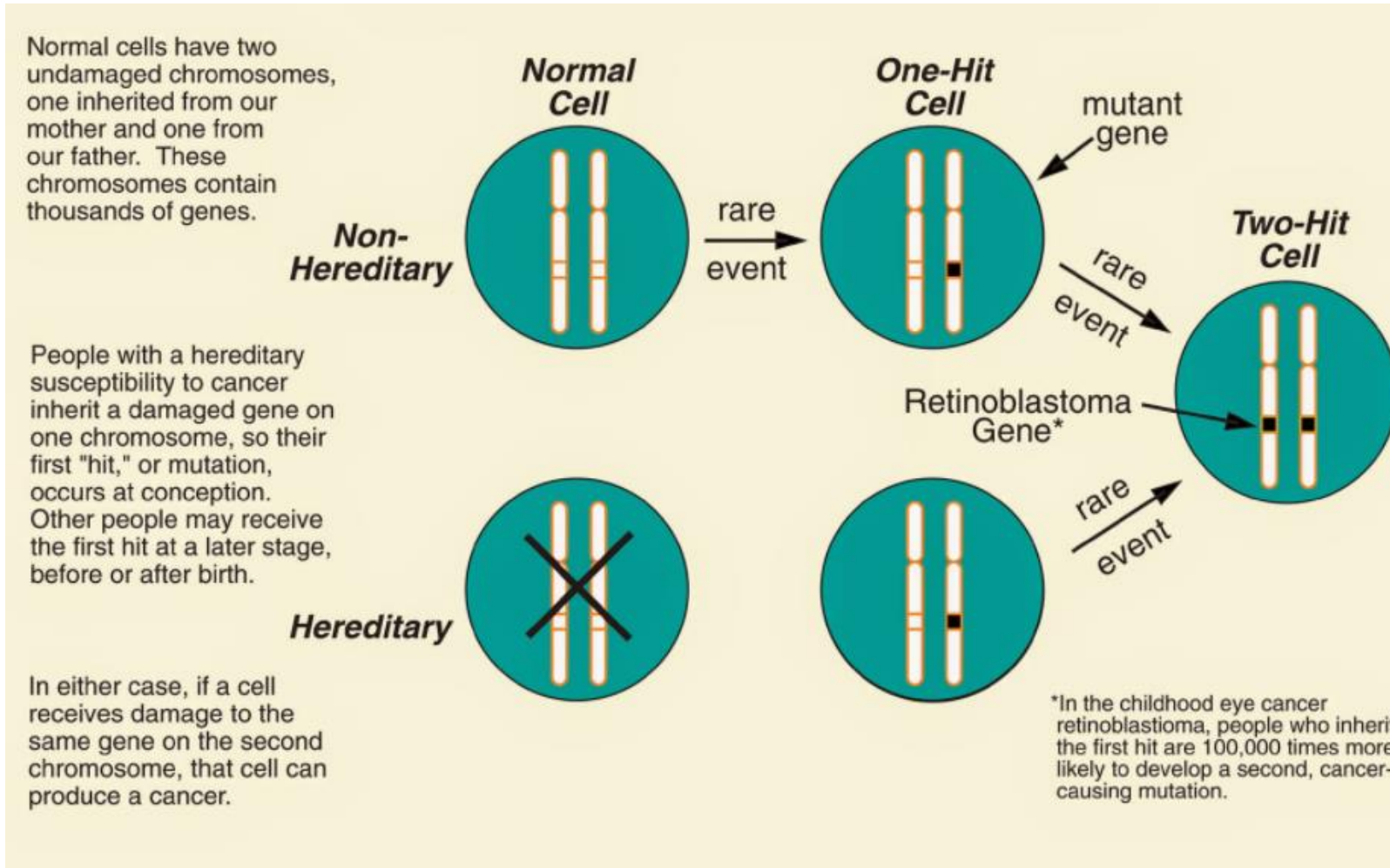
Cancer Predisposition Genes

- Oncogenes have on the cellular level a **dominant effect**.
- Loss of function of tumor suppressor genes have on cellular level a **recessive effect**.

Tow-Hit Hypothesis

The loss of both alleles of a tumor suppressor gene through 2 independent mutation events is a central component of the occurrence of cancer. In the case of familial cancer dispositions, a mutation is already constitutionally present so that only a single additional mutation event is necessary.

Two-Hit Hypothesis



Inherited Cancer Syndromes

- Familial cancer is a major component of the work of a clinical geneticist
- Common examples of an inherited cancer syndromes:
 1. HNPCC (Lynch syndrome)
 2. Familial adenomatous polyposis (FAP)
 3. Breast cancer
 4. Prostate cancer

Lynch Syndrome

- Also known as hereditary **non-polyposis colorectal** cancer (HNPCC).
- Incidence is around 1:1000 (**5%** of all colorectal cancers),
- The most important characteristic of Lynch syndrome is the family-like incidence of colon cancer or certain other cancers at a relatively young age: the median age is 45–60 years old.
- Often ($> 50\%$) of the cases are to be found in the right section of the colon.

Lynch Syndrome

- The risk of developing a colorectal cancer in the course of life with HNPCC is around **70%** in male carriers of Lynch syndrome mutation and in female up to **50%**.
- In addition, women have an additional risk up to **60% for endometrial** cancer.
- The more frequent extracolonic neoplasms also include:
Stomach cancer (lifetime risk up to 13%), Ovaries (12%), small intestine (6%), biliary tract (4%), Urothelium (4%), CNS (2%) and pancreas.

Genetic and Aetiology of Lynch Syndrome

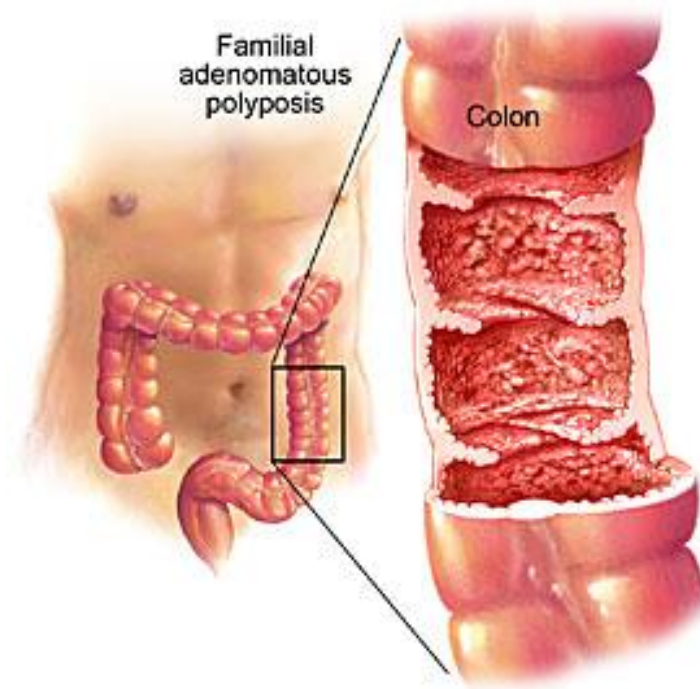
- HNPCC is usually inherited as an **autosomal dominant** pattern.
- The cause is mutation in genes of the **DNA mismatch repair (MMR)** systems. Most often in the genes *MLH1* or *MSH2*, which together make up about **90%** of the cases.
- Other Lynch Syndrome Associated genes are *MSH6*, *PMS2* and *EPCAM* gene.

Familial adenomatous polyposis (FAP)

- 1%
- Is an inherited disorder affected individuals develop numerous (thousands) polyps of the large bowel, which can involve its entirety.
- People with the classical type of FAP may begin to develop multiple noncancerous (benign) growths (polyps) in the colon as early as their childhood years.

Familial adenomatous polyposis (FAP)

- There is a high risk of carcinomatous change taking place in these polyps, with more than 90% of persons with FAP eventually developing bowel cancer by age of 50 years.



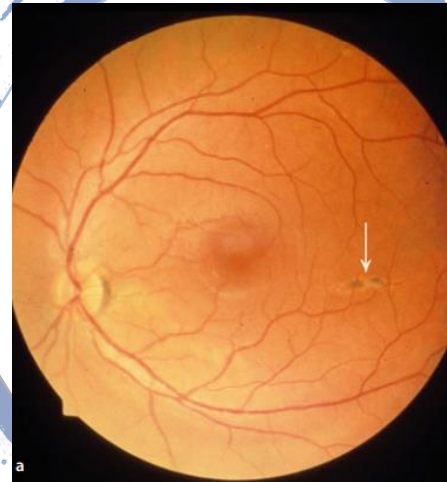
- **Familial adenomatous polyposis can have different inheritance patterns:**
- When familial adenomatous polyposis results from mutations in the *APC* gene, it is inherited in an **autosomal dominant** pattern.
- When familial adenomatous polyposis results from mutations in the *MUTYH* gene, it is inherited in an **autosomal recessive pattern**.

Extracolonic manifestations of FAP

- Congenital hypertrophy of the retinal pigment epithelium (CHRPE) with around 85% of the cases; **a.**
- Osteoma of the mandibula and the skull (90%).
- Epidermoid cysts; **b.**
- Desmoid (5-10%) which is **the most common** cause of death of colectomised patients with FAP; **c.**

- **Increased risk of the following tumors:**

Papillary Thyroid cancer, medulloblastoma of the brain (the association of FAP with CNS tumors is called **Turcot's syndrome**), sarcomas and hepatoblastomas.



A large, flowing pink ribbon graphic that forms a stylized knot or bow, positioned on the right side of the image. The ribbon has a slight gradient and a soft shadow, giving it a three-dimensional appearance.

Breast
CANCER

Breast Cancer

- This is **the most common** cancer in women.
- Approximately 1 in 8-9 women can develop the disease in their lifetime.

- **Risk** to a female relative is greater when one or more of the following factors is present:

- 1- Clustering of cases in close female relatives.
- 2- Early age (<50 years) at presentation.
- 3- Occurrence of bilateral disease.
- 4- Additional occurrence of ovarian cancer.
- 5- Paternal (or close male relative) history of breast cancer.

- Molecular studies of breast cancer tumors have revealed that breast cancer susceptibility genes ***BRCA1*** and ***BRCA2*** are frequently implicated.

***BRCA1* and *BRCA2* Genes**

- Family studies of early-onset or premenopausal breast cancer showed that it behaved like **a dominant trait in** many families.
- Pathogenic variants in *BRCA1*&2 account for about **15%** of familial breast cancer cases.
- Linkage analysis then mapped a locus to chromosome **17q**, eventually leading to identification of the ***BRCA1*** gene.
- A proportion of families with early-onset breast cancer that did not show linkage to this region showed linkage to chromosome **13q**, resulting in the identification of the ***BRCA2*** gene.

Mutations in *BRCA1* and *BRCA2*

- Account for **40% to 50%** of familial breast cancer.
- Pathogenic variants in *BRCA1* and *BRCA2* account for about 15% of the cases.
- Carriers of pathogenic *BRCA1* variants have a **60% to 90%** lifetime risk of developing the disease, as well as **a 40% to 60%** of ovarian cancer.
- The lifetime cancer risks for *BRCA2* gene mutation carriers are similar, with a breast cancer risk of **45% to 85%** and a slightly lower ovarian cancer risk of **10% to 30%**.

- Bearing in mind the population breast cancer risk of 1 in 8, and ovarian cancer risk of 1 in 50, the genes clearly carry a significantly elevated risk.
- In addition, **male *BRCA2* gene carriers** have lifetime prostate cancer risk of around **25%**. Male breast cancer risk is elevated in carriers of both *BRCA1* and *BRCA2* mutations, although it is higher in *BRCA2* gene carriers.

How Can you Prevent Genetic Cancers?

Consider the following tips to reduce your risk:

- Don't use tobacco.
- Eat a healthy diet.
- Maintain a healthy weight and be physically active.
- Get vaccinated.
- Avoid risky behaviors.
- Predictive test if there is known familial mutation in the family.
- Get regular medical care (screening programs).

Genetic Counseling in Familial Cancer

- Recognition of individuals with an inherited susceptibility to cancer usually relies on taking a careful family history to document the presence or absence of other family members with similar or related cancers.
- There are several other features that can suggest an inherited cancer susceptibility syndrome in a family.

Features Suggestive of an Inherited Cancer Susceptibility Syndrome in a Family

- Several close (first- or second-degree) relatives with a common cancer
- Several close relatives with related cancers (e.g., breast and ovary or bowel and endometrial)
- Two family members with the same rare cancer
- An unusually early age of onset
- Bilateral tumors in paired organs
- Tumors in two different organ systems in one individual

Screening for Familial Cancer

- Refer to the textbook page 203-208

- The test should detect a malignant or premalignant condition at a stage before its producing symptoms, with high sensitivity and specificity
- The treatment of persons detected by screening should improve the prognosis
- The test should preferably be non-invasive
- Adequate provision for prescreening counseling and follow-up should be available

Any Questions?

